

Exercise and worked solution

Simulation & machine learning

Four parts to simulate and learn from data: the Monte Carlo error, the bootstrap and VaR, data preparation for ML, and evaluating a classifier. Work through each part before reading the solution.

Part A · The Monte Carlo error

PROBLEM

A Monte Carlo simulation with $b = 8,000$ replications gives a standard error of 0.60. (a) How many replications to bring the standard error to 0.15? (b) Why does precision improve so slowly?

SOLUTION

- (a) We want to divide the error by 4 ($0.60 \rightarrow 0.15$). Since $\text{s.e.} \propto 1/\sqrt{b}$, we must multiply b by $4^2 = 16$, i.e. **128,000** replications.
- (b) The CLT gives $\text{s.e.} = \sigma/\sqrt{b}$: the error decays as $1/\sqrt{b}$, not $1/b$. Halving the error thus requires $4\times$ more replications, and a factor 10 of precision needs $100\times$. This is the square-root cost, essential for sizing a simulation.

Part B · Bootstrap and VaR

PROBLEM

A risk manager estimates an annual S&P 500 VaR by bootstrap. (a) Draw with or without replacement, and why are repeats not an error? (b) The returns show volatility clustering. iid bootstrap or CBB, and what block size? (c) What fundamental limit stops the bootstrap from forecasting a novel crisis?

SOLUTION

- (a) **With replacement**: the same observation can recur several times, which is normal (at least one repeat is near-certain once you draw many samples). This draw-with-replacement is what lets us build many samples from a single dataset.
- (b) A **CBB** (circular block bootstrap), because the data are dependent: it draws blocks of q consecutive observations and preserves the volatility clusters. Block size $q \approx \sqrt{n}$. An iid bootstrap would break the clusters and underestimate the extreme VaR.
- (c) The bootstrap resamples history: it can **never generate a loss worse than the largest already observed** (Black Swan). It also mixes all regimes, making it misleading when the market enters an abnormal state. A good simulation model can extrapolate beyond history.

Part C · Preparing data for ML

PROBLEM

(a) A feature has a mean of 40 and a standard deviation of 5. Standardize the observation $x = 52$. (b) When is standardization preferred over min-max normalization? (c) Do we also rescale the target?

SOLUTION

- (a) $\tilde{x} = (x - \mu)/\sigma = (52 - 40)/5 = 12/5 = \mathbf{2.4}$: the observation is 2.4 standard deviations above the mean. After standardization, the feature has mean 0 and variance 1.
- (b) Standardization is preferred **with outliers**: min-max normalization (division by $\max - \min$) would then squeeze most observations into a narrow range, unrepresentative of their true spread.
- (c) **No**: we rescale only the **features** (inputs), never the **target** (the variable to predict). Rescaling puts features on the same scale so distances and penalties compare them fairly.

Part D · Evaluating a classifier

PROBLEM

A fraud-detection model gives, on 500 transactions: TP = 180, FN = 60, FP = 40, TN = 220. (a) Compute accuracy, precision and recall. (b) Is an AUC of 0.5 an “average” score? (c) An undetected fraud is very costly: how to adjust the threshold Z, and at what cost?

SOLUTION

1. (a) Accuracy = $(180+220)/500 = \mathbf{80.0\%}$. Precision = $TP/(TP+FP) = 180/220 = \mathbf{81.8\%}$. Recall = $TP/(TP+FN) = 180/240 = \mathbf{75.0\%}$.
2. (b) **No**: an AUC of 0.5 is the chance diagonal, i.e. **no predictive power**. AUC = 1 is a perfect classifier; an AUC below 0.5 is worse than chance.
3. (c) A false negative (undetected fraud) is costly: we **lower the threshold Z** (e.g. to 0.2 instead of 0.5) to predict “fraud” whenever even a small risk exists. This raises recall (fewer missed frauds), at the cost of **lower precision** (more false alarms): this is the trade-off traced by the ROC curve.

Key takeaways: the Monte Carlo error decays as $1/\sqrt{b}$ ($\div 2 \Rightarrow \times 4$ replications); the bootstrap draws with replacement and never creates a novel loss (CBB if dependence, $\sqrt{n}$ blocks); rescale the features (not the target), standardization with outliers; and a classifier is judged by precision, recall and AUC (0.5 = no predictive power), the threshold Z trading off false positives against false negatives.